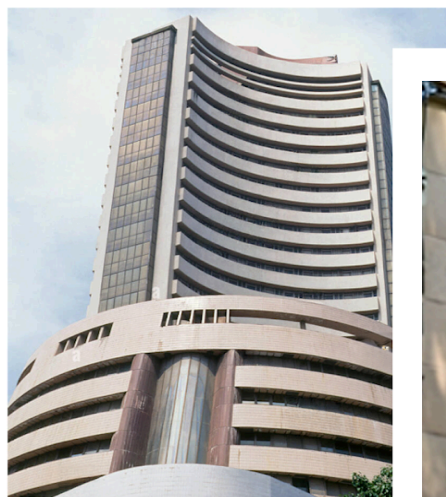


General Knowledge

Indian Economy

STUDY NOTES

Indian Agriculture

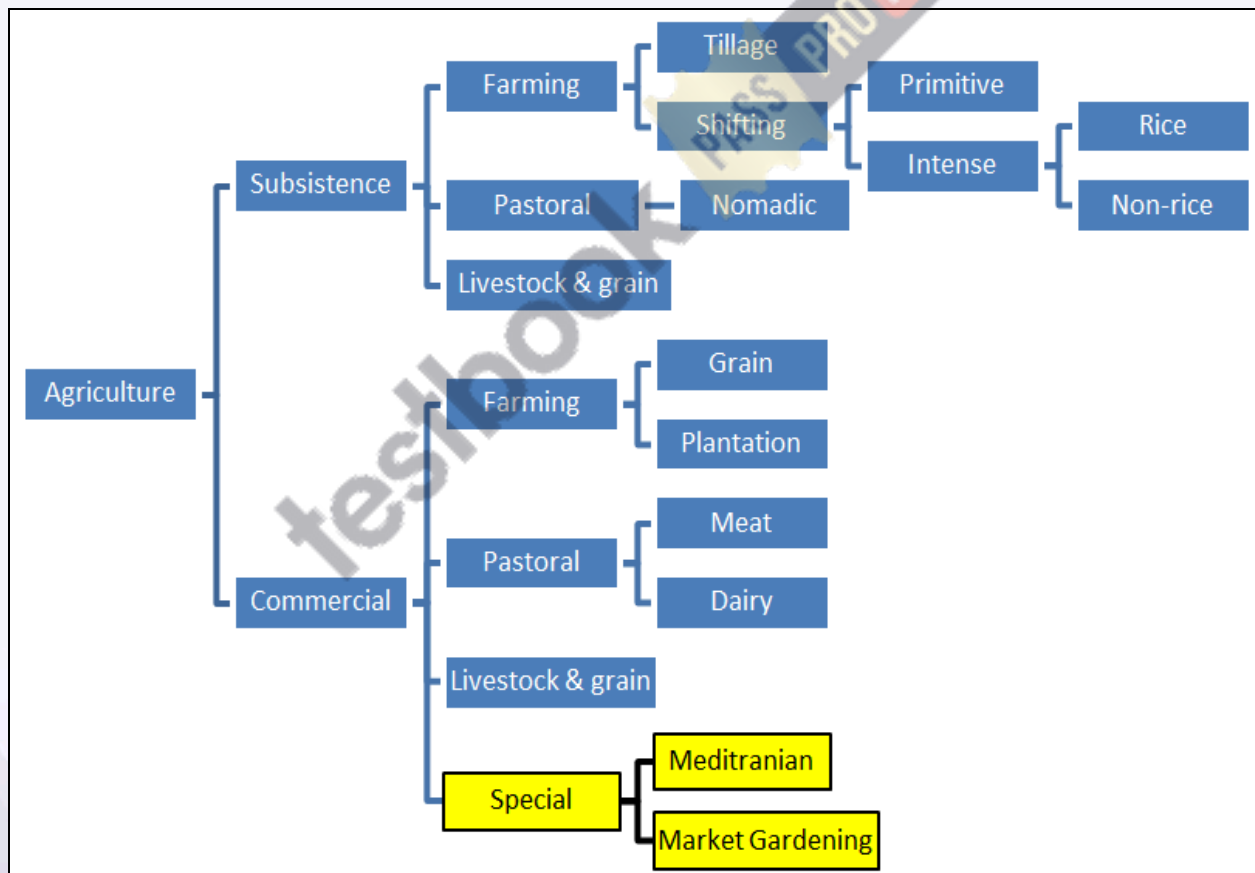


INDIAN AGRICULTURE

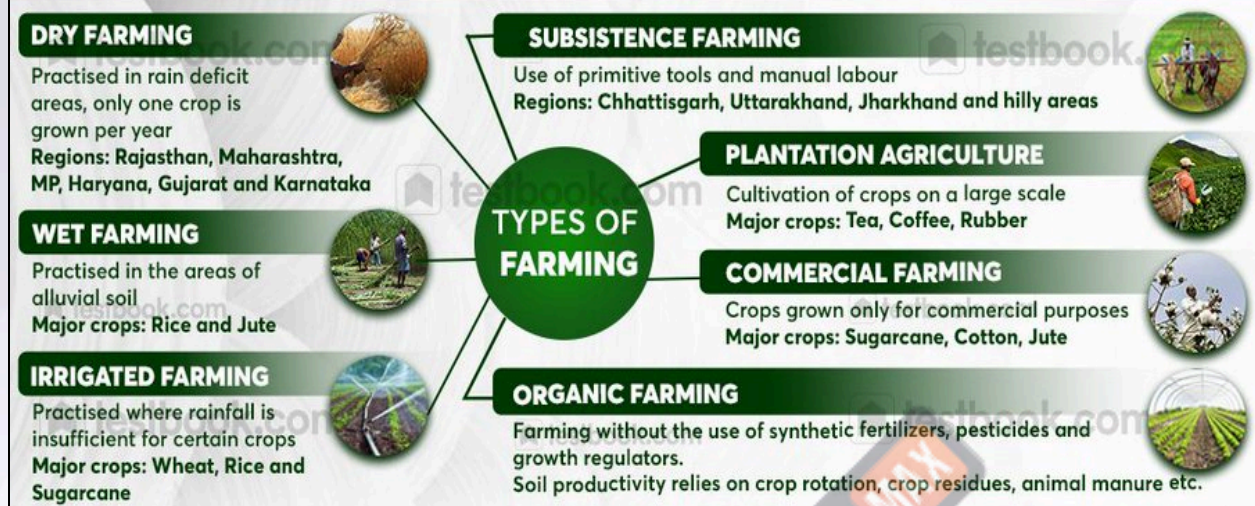
Agriculture encompasses farming, soil cultivation, crop cultivation, and animal husbandry for food, wool, and various products. It remains the **primary industry in India**, covering **43% of the country's geographical area**.

Over **58% of rural households** rely on agriculture as their primary livelihood. The agricultural growth rate has been consistent at **4.6% annually** over six years but declined to **3% in 2021-22**, as per the **Economic Survey 2022-23**. The **GDP contribution of agriculture** was **19.9% in 2020-21**, **19% in 2021-22**, and **18.3% in 2022-23**.

Types of Agriculture:



IMPORTANT TYPES OF FARMING IN INDIA



Type of Culture	Description
Agriculture	The practice of cultivating land and growing crops for food and raw materials.
Horticulture	The art and science of growing fruits, vegetables, flowers, and ornamental plants.
Floriculture	The cultivation of flowers and decorative plants for commercial use or personal enjoyment.
Animal Husbandry	The practice of breeding and caring for livestock, including cattle, goats, sheep, and poultry.
Aviculture	The raising and care of birds, especially for the purpose of breeding and sale.
Aquaculture	The cultivation of aquatic organisms like fish, shellfish, and algae in controlled environments.
Apiculture	The practice of beekeeping for honey, wax, and pollination services.
Sericulture	The production of silk by rearing silkworms and harvesting their cocoons.

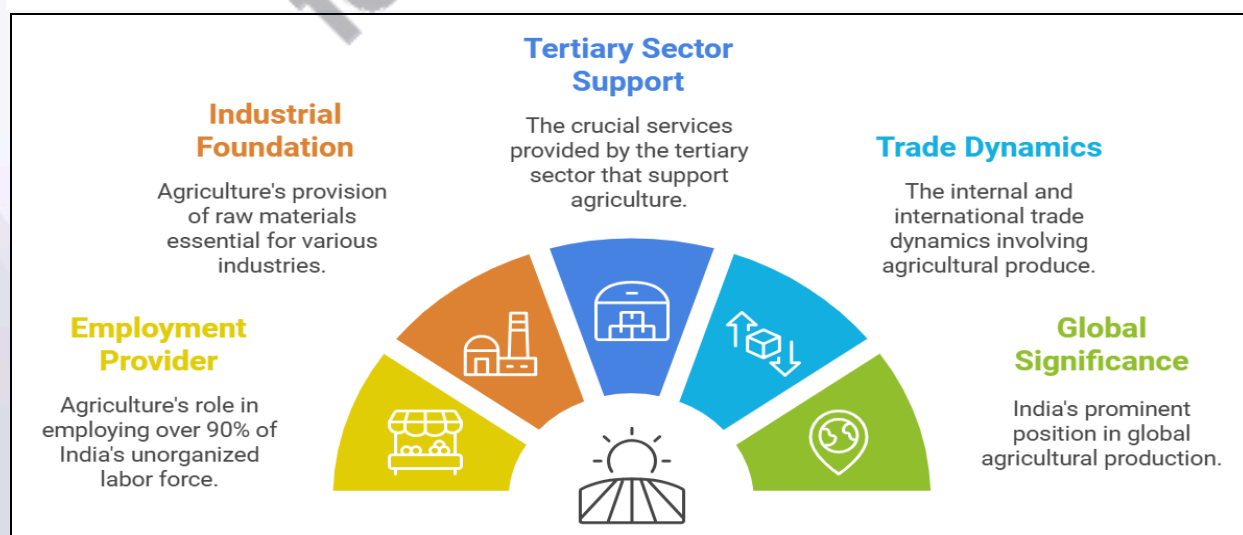
Viticulture	The cultivation and harvesting of grapes, particularly for wine production.
Fungi Culture	The cultivation of fungi such as mushrooms for food, medicine, or other uses.
Arboriculture	The cultivation and management of trees and shrubs for timber or aesthetic purposes.
Vermiculture	The practice of breeding and rearing earthworms to enhance soil fertility and composting.
Moriculture	The cultivation of mulberry plants, primarily for silkworm rearing in sericulture.
Pisciculture	The breeding and raising of fish in controlled environments for food production or conservation.

Indian Cropping Pattern with Seasons

Season	Sowing Period	Harvesting Period	Major Crops	Examples	Details
Rabi	October to December	March to May	Wheat, Barley, Gram, Mustard, Peas	Wheat: Punjab, Haryana Mustard: Rajasthan Gram: Madhya Pradesh	- Requires cool weather for growth and warm weather for maturity - Grown in winter - Relies on irrigation or winter rainfall.

Kharif	June to July	September to October	Rice, Maize, Cotton, Bajra, Jowar, Groundnut, Soybean	Rice: West Bengal, Tamil Nadu Cotton: Gujarat, Maharashtra Maize: Bihar	- Requires high temperature and substantial rainfall - Grown in monsoon - Success heavily depends on monsoon consistency.
Zaid	March to June	June to August	Watermelon, Muskmelon, Cucumber, Fodder crops, Sugarcane	Watermelon: Uttar Pradesh Cucumber: Rajasthan Sugarcane: Uttar Pradesh	- Short-duration crops grown during summer - Requires warm weather and longer day length - Grown mainly where irrigation facilities are ample.

Key Aspects of Indian Agriculture



Agriculture in Five-Year Plans

Plan	Description	Agricultural Growth Rate
First Plan (1951-56)	Focused on solving the food crisis; 14% of the plan outlay was allocated to agriculture.	2.71%
Second Plan (1956-61)	Reduced agricultural allocation to 11.7%; focused on industrial development.	3.17%
Third Plan (1961-66)	Introduced IADP and HYVP; set ambitious targets but saw negative growth.	-0.73%
Fourth Plan (1969-74)	Emphasized science and technology in agriculture.	2.57%
Fifth Plan (1974-79)	Actual foodgrain production exceeded targets.	3.28%
Sixth Plan (1980-85)	Witnessed the Second Green Revolution in 1983-84.	4.3%
Seventh Plan (1985-90)	Allocated 6% to agriculture; failed to meet targets for cotton and other sectors.	3.47%
Eighth Plan (1992-97)	Achieved robust growth due to favourable weather conditions.	4.68%
Ninth Plan (1997-2002)	Struggled to meet targets.	2.44%
Tenth Plan (2002-07)	Introduced the National Agricultural Policy for Sustainable Growth.	2.3%
Eleventh Plan (2007-12)	Fell short of the 4% target for agricultural growth.	3.6%

Twelfth Plan (2012-17)	Confronted challenges like climate change and market uncertainties.	Varied growth rates
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Agricultural Growth (2017-2023)

Year	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23
Growth Rate (%)	6.6%	2.1%	5.5%	3.3%	3.0%	3.5%

Agricultural Inputs and Practices

- **Natural Farming:**
 - Promotes **chemical-free farming** practices.
 - Restores soil fertility and requires less water.
 - Supported under the **Bharatiya Prakritik Krishi Paddhati Programme (BPKP)**.
- **Organic Farming:**
 - Utilizes crop rotation, green manure, and compost for sustainable farming.
 - Focuses on **quality control, soil health, and market development**.
- **Seeds:**
 - **National Seeds Policy, 2002** addresses the low seed replacement rate in India.
 - **Seed Bank Initiative** ensures availability during unforeseen needs.
- **Fertilizer Subsidies:**
 - The government provides affordable fertilizers through the **Nutrient-Based Subsidy Scheme**.
 - The **New Urea Policy (2015)** promotes energy-efficient indigenous production.
- **Irrigation:**
 - **Sprinkler Irrigation:** Suitable for high-density crops.
 - **Drip Irrigation:** Ensures water and fertilizer efficiency.

Agricultural Schemes

Scheme	Year	Target Group	Features
Atal Pension Yojana (APY)	2015	Unorganized sector workers	Guaranteed pension of ₹1,000-₹5,000 based on contributions.

PM Kisan Samman Nidhi Yojana	2018	Small and marginal farmers	Provides ₹6,000 annually in three instalments.
Pradhan Mantri Fasal Bima Yojana	2016	Farmers	Insurance for crop loss due to natural calamities.
PM Kisan Maan Dhan Yojana	2019	Small farmers	Monthly pension of ₹3,000 after 60 years of age.
Soil Health Card Scheme	2015	All farmers	Distribute soil health cards to ensure sustainable farming practices.
KUSUM Scheme	2019	Farmers	Promotes solar-powered irrigation systems with subsidies.
Pradhan Mantri Krishi Sinchai Yojana (PMKSY)	2015	Farmers	Expands irrigation potential and promotes water-efficient technologies.
National Food Security Mission (NFSM)	2007	Farmers	Enhances food grain production through area expansion and productivity improvements.

Agriculture Reforms

Reform/Policy	Key Features
National Agricultural Commission (2004)	Proposed an Agricultural Renewal Plan with components such as soil health enhancement, irrigation management, credit and insurance facilities, technological reforms, and assured marketing.
National Agriculture Policy (2007)	Focused on the economic well-being of farmers , new technologies like biotechnology, effective MSP implementation , and expanding the food security basket with nutritious millets.

Minimum Support Prices (MSP)	Ensures prices paid to farmers cannot fall below the announced MSP, stabilizing farmer incomes.
Procurement Prices	Higher than MSP; enables government to buy crops directly from farmers to ensure market stability.
Central Issue Prices (CIP)	Subsidized rates for food grains under the Public Distribution System (PDS) for Below Poverty Line families.
Support through MSP	Focused on specific crops like rice and wheat; CACP recommends MSP for 23 commodities.
Public Distribution System (PDS)	Jointly managed by Central and State Governments , distributing essential commodities like wheat, rice, and sugar.
Targeted Public Distribution System (TPDS)	Introduced a two-tier pricing structure for BPL and APL families , along with the Antyodaya Anna Yojana for the poorest families.
Revamped Public Distribution System (RPDS)	Launched in 1992 to strengthen PDS in remote areas, introducing expanded ration cards and additional commodities.
Fortification of Rice	Pilot scheme (2019) for fortified rice distribution to combat malnutrition across 15 districts in 11 states.
National Food Security Act (NFSA), 2013	Ensures 5 kg of food grains per person per month for priority households and 35 kg for Antyodaya households, with subsidized pricing of ₹3/kg for rice, ₹2/kg for wheat, and ₹1/kg for millets.
Monitoring and Transparency	State food commissions oversee implementation, with grievance redressal mechanisms and social audits for transparency.

Land Reforms

Land reforms aim to achieve **social equity** and **economic growth** by addressing critical issues related to land ownership and usage. The program focused on:

- **Abolition of intermediaries.**
- **Settlement and regulation of tenancy.**
- **Regulation of the size of landholdings** to ensure equitable distribution.

Post-Independence Land Reforms

- The **Zamindari system** was abolished to remove intermediaries, granting ownership or occupancy rights to actual cultivators.
- Tenancy reforms were implemented through **state-specific legislation** to protect tenant rights and improve their conditions.
- **Landholding ceilings** were introduced to prevent excessive accumulation, with around **1633 lakh hectares consolidated by 2004.**
- **Cooperative farming** was introduced to promote collective farming practices.
- The **Bhoodan Movement**, led by **Acharya Vinoba Bhave**, encouraged voluntary land donations to benefit landless farmers.
- **Vedire Ramachandra Reddy** was the first person to donate land to the poor under the Bhoodan movement, which was started by **Vinoba Bhave in 1951.**

Categorization of Land Holdings

Size of Holding	Classification
Less than 1 hectare	Marginal Holding
1-2 hectares	Small Holding
2-4 hectares	Semi-medium Holding
4-10 hectares	Medium Holding
More than 10 hectares	Large Holding

National Land Records Modernisation Programme (NLRMP)

- **NLRMP** was launched in **2008** by merging:

- **Computerisation of Land Records (CLR).**
- **Strengthening of Revenue Administration and Updating of Land Records (SRA and ULR).**
- Objectives of NLRMP:
 - Modernizing **land record management**.
 - Reducing **land disputes** through better transparency.
 - Moving towards **guaranteed conclusive titles** for immovable properties.

Digital India Land Records Modernisation Programme (DILRMP)

- Launched in **August 2008**, DILRMP aims to:
 - **Computerize land records**, including mutation and settlement records.
 - Digitize maps and surveys to enhance transparency and reduce disputes.
 - Improve the accessibility of land records and prevent property fraud.
- **Benefits of Digitization:**
 - Provides **transparent land ownership titles** monitored by government officials.
 - Expedites **land transactions** and reduces construction timelines.
 - Makes property prices more appealing by lowering development costs.

Model Agricultural Land Leasing Act, 2016

- Introduced by **NITI Aayog**, the Act aims to address issues related to agricultural tenancy laws.
- Key Features:
 - Facilitates the **leasing of agricultural land** for efficient usage.
 - Recognizes farmers on leased land for access to **institutional credit**.
 - Promotes **agricultural efficiency** and **equity**, improving both productivity and occupational mobility.

Minimum Support Price (MSP)

- **Minimum Support Price (MSP)** is a pre-determined price set by the Government of India to protect farmers against sharp falls in agricultural prices.
- MSP ensures that farmers receive a **minimum price** for their produce, safeguarding them against market volatility.

- Introduced in **1965-66**, MSP was a key component of India's agricultural pricing policy to boost production and ensure food security.

Objectives of MSP

- To **ensure fair remuneration** for farmers.
- To **stabilize agricultural incomes** by mitigating the risks associated with price fluctuations.
- To **encourage higher investment** in agriculture by assuring minimum returns.
- To act as a **price signal** for farmers, guiding them on cropping patterns and diversification.
- To ensure **food security** by incentivizing the production of essential crops.

How MSP is Determined

- MSP is recommended by the **Commission for Agricultural Costs and Prices (CACP)**, based on several criteria:
 - **Cost of production (A2+FL and C2):**
 - **A2:** Covers actual costs incurred by farmers (seeds, fertilisers, irrigation, etc.).
 - **FL:** Family labour cost.
 - **C2:** Comprehensive cost, including A2+FL plus imputed rent and interest on owned land and capital.
 - **Demand-supply dynamics** of agricultural products.
 - **Market price trends** (domestic and international).
 - **Inter-crop price parity** to ensure fairness among various crops.
 - **Terms of trade** between agricultural and non-agricultural sectors.
 - Implications for **food inflation** and consumer interests.

Crops Covered Under MSP

- MSP is announced for **23 crops**:
 - **7 Cereals:** Paddy, wheat, barley, jowar, bajra, maize, and ragi.
 - **5 Pulses:** Gram, arhar (tur), moong, urad, and lentil.
 - **7 Oilseeds:** Groundnut, rapeseed, mustard, soybean, sunflower, sesame, and niger seed.

- **4 Commercial Crops:** Cotton, sugarcane (Fair and Remunerative Price), raw jute, and copra.

Implementation of MSP

- MSP is enforced through **government procurement agencies** such as:
 - **Food Corporation of India (FCI)** for food grains.
 - **National Agricultural Cooperative Marketing Federation of India (NAFED)** for oilseeds and pulses.
 - **Cotton Corporation of India (CCI)** for cotton.
- **State Governments** and their agencies also participate in procurement under MSP.

Challenges with MSP

- **Regional Disparity:** Procurement is primarily focused on a few states like Punjab, Haryana, and Madhya Pradesh for crops like wheat and rice, leaving other regions underrepresented.
- **Limited Crop Coverage:** MSP mostly benefits farmers growing cereals and commercial crops; horticultural crops and others remain outside its scope.
- **Inefficient Procurement:** A significant proportion of small and marginal farmers are unable to access MSP benefits due to weak procurement infrastructure.
- **Overemphasis on Certain Crops:** Encourages water-intensive crops like paddy and wheat, leading to environmental issues like groundwater depletion.
- **Economic Burden:** The growing cost of subsidies due to MSP places a heavy burden on government finances.
- **Exclusion of Marginal Farmers:** Many small-scale farmers sell their produce in local markets at lower prices due to a lack of awareness or access to procurement centres.

Revolutions in Indian Agriculture

Green Revolution in India

- The **Green Revolution** was introduced in **post-1965** to increase food grain production and ensure food security in India.

- It relied on the use of **high-yielding seed varieties**, increased **fertilizer** and **irrigation** usage, and expanded farming areas.
- The revolution focused on **double-cropping** of existing farmland and using seeds with improved yields, particularly for wheat and rice.
- It made India **self-sufficient** in food grain production.
- **Drawbacks** of the Green Revolution included environmental impacts, **soil nutrient depletion**, **water resource depletion**, and higher input costs.
- Health impacts were observed due to the **extensive use of chemical fertilizers** and pesticides.

Bringing Green Revolution in Eastern India Programme (BGREI)

- The **BGREI** was launched in **2010-11** as part of the **Rashtriya Krishi Vikas Yojana** to promote agriculture in Eastern India.
- It aimed to address resource allocation and utilisation in states like **West Bengal, Assam, Bihar, Chhattisgarh, Jharkhand, Odisha, and Eastern Uttar Pradesh**.
- The program resulted in a significant increase in food grain production, with an **11.99% growth rate** during **2011-12**, surpassing the national growth rate of 2.2%.
- The program was extended beyond the **Twelfth Five-Year Plan (2015-16 to 2019-20)** as a centrally sponsored scheme.
- **Key benefits** included improved productivity and efficient resource utilisation in the targeted regions.

Second Green Revolution in India

- The **Second Green Revolution** focuses on **sustainability** and enhancing agricultural productivity.
- It emphasises **increasing crop yields** in Eastern states through advanced farming techniques.
- The revolution promotes **organic farming, contract farming**, and the amendment of the **Agricultural Produce Marketing Committee (APMC) Acts**.
- It includes investments in **research** to develop drought-proof crops and combat climate change.
- It focuses on building **supply chains**, and **cold chains**, and improving soil quality through innovative agricultural practices.

- Other initiatives include **rainwater harvesting**, **watershed development**, and enhancing credit availability for farmers.

Major Agricultural Revolutions

Revolution Name	Focus
Green Revolution	Cereals, wheat, and legumes
White Revolution	Milk and dairy products
Silver Revolution	Eggs and poultry
Yellow Revolution	Edible oil production
Blue Revolution	Fisheries
Pink Revolution	Prawns and meat processing
Golden Revolution	Honey production
Golden Fibre Revolution	Jute production
Silver Fibre Revolution	Cotton production

Other Notable Agricultural Revolutions

Revolution Name	Description
Evergreen Revolution	The concept by Dr. MS Swaminathan; focuses on sustainable organic and green farming.
Rainbow Revolution	Announced in 2000; aimed at 4% agricultural growth with multiple reforms.
Tricolour Revolution	Coined by PM Narendra Modi; includes Saffron Energy, White, and Blue Revolutions .

Operation Green	Announced by Finance Minister Arun Jaitley; promotes tomato, onion, and potato production.
Sweet Revolution	Promotes scientific beekeeping under the National Beekeeping and Honey Mission.

India's Position in World Agriculture Production

Produce	Global Rank (2024)
Wheat	2nd
Rice/Paddy	2nd
Total Pulses	1st
Groundnut (in shell)	2nd
Vegetables (including melons)	2nd
Fruits (excluding melons)	2nd
Sugarcane	2nd
Tea	2nd
Jute and Jute-like fibres	1st
Cotton (lint)	2nd
Total Milk	1st
Total Eggs	3rd
Total Meat	5th